Hardening Multi-Agent Personal Health Systems Across the Inpatient-to-Home Continuum via 4D Time-Locked EHR States, Fail-Closed Target Trials, and Apple Vision Pro (`visionOS`) Spatial Decision Crossroads

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https://github.com/deesatzed/personal-health-agent-bridge
https://github.com/deesatzed/spatial-clinical-router

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Abstract

Consumer-facing multi-agent personal health systems—most notably the Personal Health Agent (PHA) architecture introduced by Heydari, Xu, et al. (*Nature*, 2026; `doi:10.1038/s41586-026-11161-8`), which coordinates a **Data Science Agent (DSA)**, **Domain Expert Agent (DEA)**, and **Health Coach Agent (HCA)**—have demonstrated substantial advantages over monolithic large language models (LLMs) in longitudinal wearable analytics, clinical reasoning, and motivational interviewing. However, when deployed across the high-risk **inpatient-to-home post-discharge transition** (Days 1–90 following acute hospitalization), consumer multi-agent health systems encounter four structural clinical safety hazards: (1) **asynchronous outpatient timestamp look-ahead leakage**, where joining continuous wearable streams (t_{sync}) with outpatient electronic health record (EHR) laboratories by specimen collection time (t_{chart}) rather than patient-portal verification time (t_{store}) leaks delayed results; (2) **implicit negative imputation (`Unavailable == Negative`)**, where unordered outpatient perfusion or natriuresis workups are treated as normal findings; (3) **observational healthy-adherer and acute-flare confounding**, where the DSA infers causal benefits of exercise or fluid adjustments across non-overlapping clinical states; and (4) **unbounded autonomous escalation or alert fatigue** from uncalibrated wearable syncs. Conversely, acute inpatient decision architectures such as the **4-Stage Spatial Clinical Router (`SALVAGE-AI`)** terminate at hospital discharge, leaving a critical monitoring blind spot during outpatient recovery.

Here we present a **bidirectional architectural synthesis (`PHA` × `SALVAGE-AI`)** that bridges acute inpatient decision support and home recovery across two high-acuity post-discharge cohorts ($N = 160$ synthetic benchmark trajectories: $n = 80$ post-diabetic foot infection limb salvage [`POST_DFI_SALVAGE`] and $n = 80$ post-sepsis cardiorenal recovery [`POST_SEPSIS_CARDIORENAL`]). First (`SALVAGE-AI` → `PHA`), we harden the 3-agent PHA with (i) a 4D `storetime` state lock ($t_{\text{store}} \leq t_k$), eliminating a +0.1890 **optimistic AUROC inflation** (0.8110 strict `storetime` vs. 1.0000 leaky `charttime`) at Day 7 post-discharge; (ii) a *Drosophila* Kenyon-Cell (`240-KC`, 95.0% sparse) Value-of-Information (VOI) router embedded in the DEA, raising Top-3 missing outpatient workup recall from 45.62% (monolithic LLM) and 68.75% (unguarded 3-agent PHA) to 100.00%; (iii) a fail-closed Hernán–Robins target-trial gate in the DSA that blocks 100.0% of unidentifiable behavioral/diuretic causal claims during acute physiological flares; and (iv) an **Apple Vision Pro (`visionOS 27.0 / iPadOS 27.0`) Spatial Decision Crossroads & `M0-M5` Symbiosis Gate** (90.0 Hz, 0.68 ms, hard clinical safety caps Autonomy $\leq 40\%$ and Deference $\geq 80\%$) that suppresses 100.0% of routine benign wearable syncs from interrupting clinicians while maintaining 100.0% wake sensitivity on clinical red flags. Second (`PHA` → `SALVAGE-AI`), the 3-agent PHA extends inpatient anti-reward-hacking utility functions across the 90-day home continuum, delivering structured **Motivational Interviewing (`OARS`)** on stable recovery days while automatically halting autonomous lifestyle coaching (0.0% **unsafe coaching rate** vs. 12.5% in unguarded PHA and 34.0% in a monolithic LLM) to mint an HMAC-SHA256 signed `sdr.decision.v1` Clinician `NowCard` upon decompensation.

1 Introduction: The Post-Discharge Continuum Gap

Recent work by Heydari, Xu, et al. [1] established a foundational blueprint for consumer personal health AI by

decomposing a monolithic assistant into a collaborative multi-agent system:

1. **Data Science Agent (DSA):** Formulates statistical analysis plans and executes code over longitudinal wearable time series (resting heart rate, heart rate variability, sleep architecture, physical activity) contextualized against population percentiles and personal baselines.

2. **Domain Expert Agent (DEA):** Synthesizes personalized physiological data with authoritative clinical guidelines and medical domain literature.

3. **Health Coach Agent (HCA):** Engages users via structured behavioral psychology frameworks—specifically **Motivational Interviewing (`OARS`: Open Questions, Affirmations, Reflections, Summaries)** [2] and SMART goal setting—while an **Orchestrator** coordinates turn-by-turn agent synthesis.

While this three-specialist anatomy excels in general wellness and chronic lifestyle coaching, a major clinical translation opportunity lies in the **post-discharge hospital-to-home transition** (Days 1–90 following acute inpatient care). Nearly 20–25% of patients hospitalized for diabetic foot infection (DFI) limb salvage [3, 4] or septic cardiorenal syndrome [5, 6] experience unplanned 30-day readmission or relapse. During this window, patients generate both **high-frequency consumer/medical wearable streams** (smartwatch photoplethysmography, Bluetooth weight scales, smart offloading boot adherence, and plantar thermometry mats [7]) and **asynchronous outpatient EHR events** (send-out wound cultures, outpatient basic metabolic panels [BMP], C-reactive protein [CRP], and vascular perfusion studies).

Deploying a consumer PHA across this post-discharge transition without acute clinical safeguards exposes four failure modes:

1. **The Outpatient `charttime` vs. `storetime` Temporal Trap:** When a patient visits an outpatient phlebotomy or wound clinic at $t = 164$h post-discharge, the specimen collection timestamp ($t_{\text{chart}} = 164$h) is recorded immediately, whereas serum CRP or BMP verification ($t_{\text{store}} = 172$h) and aerobic/anaerobic wound culture speciation ($t_{\text{store}} = 212$h) post to the patient portal hours to days later [8, 9]. Retrospective evaluations that join wearable syncs at $t_k = 168$h against EHR tables using $t_{\text{chart}} \leq t_k$ leak future diagnostic results into the agent's context.

2. **The `Unavailable == Negative` Epistemic Fallacy in Outpatient Care:** Post-discharge patients frequently lack key baseline outpatient measurements—such as an unmeasured hallux toe pressure (`pedal_doppler_toe_pressure_mmhg`) in a patient with medial arterial calcinosis (ABI > 1.30), or an unordered spot urine sodium (`spot_urine_sodium_mmol_l`) in a post-sepsis patient on oral loop diuretics. Standard RAG and multi-agent prompts treat absent lab keys as normal/unremarkable rather than quantifying the **Value of Information (VOI)** of ordering the missing test.

3. **Observational Healthy-Adherer Confounding and Acute-Flare Positivity Violations:** When the DSA analyzes observational wearable logs, days with higher step counts or aggressive fluid restriction correlate strongly with lower baseline inflammation (healthy-adherer bias). Conversely, during an acute plantar thermal flare ($\Delta T_{\text{plantar}} \geq 2.2°\text{C}$) or congestive weight surge ($\Delta\text{Weight}_{72\text{h}} \geq +2.0$ kg), encouraging graded step progression or autonomous diuretic escalation violates clinical positivity [10, 7].

4. **Unbounded Autonomous Escalation vs. Clinician Alert Fatigue:** Executing an uncalibrated multi-agent loop (`DSA` + `DEA` + `HCA`) without spatial human-in-the-middle decision gating either floods clinicians with benign routine syncs or permits unsafe autonomous medical actions.

Conversely, our inpatient **4-Stage Spatial Clinical Router (`SALVAGE-AI`)** [9]—which introduced 4D `storetime` state locking, *Drosophila* `240-KC` sparse associative VOI routing [11], fail-closed Hernán–Robins target-trial gates [10, 12], and an Apple Vision Pro (`visionOS 27.0`) Spatial Decision Crossroads & `M0-M5` Symbiosis Gate for inpatient `4DMFEET` and `4D-SEPSIS` triage—previously terminated at the hospital door.

In this work, we introduce and evaluate the **Bidirectional `PHA` × `SALVAGE-AI` Bridge**, unifying Google's 3-agent Personal Health Agent [1] with the 4-stage `SALVAGE-AI` safety architecture [9].

2 Bidirectional Architectural Formulation

2.1 Direction A (`SALVAGE-AI` → `PHA`): Four Clinical Safety Guards for Multi-Agent Personal Health

Stage A1: 4D Time-Locked Multimodal Outpatient + Wearable State (`models.py`). At any post-discharge evaluation landmark t_k (hours since index hospital discharge), the patient state is represented by a cryptographic 4D state object $\mathcal{S}(t_k)$ (`TimeLockedPHALandmarkState`) that strictly separates synchronous wearable telemetry from asynchronous outpatient EHR records:

$$\mathcal{E}_{\text{visible}}(t_k) = \{e_i \in \mathcal{E}_{\text{outpatient}} \mid t_{\text{store}}(e_i) \leq t_k\} \quad (1)$$

Any outpatient lab or microbiology specimen drawn prior to t_k but verified after t_k ($t_{\text{chart}}(e_i) \leq t_k < t_{\text{store}}(e_i)$) is excluded from the feature vector and tagged as `PENDING_AT_LANDMARK`. Any clinical feature never ordered in the post-discharge window is assigned an explicit binary indicator $\text{MISSING_IND}_j = 1$ and tagged as `MISSING_UNORDERED` (`Unavailable != Negative`). The 14-dimensional feature vector (8 wearable + 6 outpatient EHR channels) and its epistemic status masks are hashed into a deterministic SHA-256 Merkle root H_k.

Stage A2: Apple Vision Pro (`visionOS 27.0`) Spatial Crossroads & `M0-M5` Symbiosis Gate (`vision_pro_spatial_gate.py`). Each 4D state $\mathcal{S}(t_k)$ is evaluated through the **Apple Vision Pro (`visionOS 27.0 / iPadOS 27.0`) Spatial Decision Crossroads & `M0-M5` Symbiosis Gate** (90.0 Hz frame budget, 0.68 ms gate evaluation latency, 95.0% **sparsity** [11]):

- **Red-Flag Hard Escalation vs. Routine Sync Suppression:** If deterministic safety rules detect a plantar thermal flare ($\Delta T_{\text{plantar}} \geq 2.2°\text{C}$), suboptimal offloading ($< 70\%$ boot adherence), acute congestive weight gain ($\Delta\text{Weight}_{72\text{h}} \geq +2.0$ kg), or cardiorenal sympathetic decompensation (RHR $\geq$ 95 bpm or HRV $\leq$ 22 ms), the Spatial Gate routes the episode to `NOW_CARD_HARD_ESCALATION`. On routine stable recovery days, the gate suppresses clinician interrupts (`SUPPRESSED_ROUTINE_SYNC`, 100.0% **benign sync suppression**).
- **`M0-M5` Symbiosis Decision-Model Gating:** Enforces (`M0`) pop-out `OptionRing` lifecycle (`Trial` $\rightarrow$ `Promoted` $\star$ $\rightarrow$ `Dormant` $\varnothing$), (`M1`) 4-state epistemic field reliability chips (`R/N/W/D`) + `If-Wrong` risk intervals, (`M2`) Earned-Autonomy Dials with hard clinical safety caps (**Autonomy** $\leq$ 40%, **Deference** $\geq$ 80%), (`M3`) Blinded Spot-Checks (`Your Call First`), (`M4`) `PrincipalCard` category promotion, and (`M5`) cryptographic lineage.

Stage A3: Fail-Closed Hernán–Robins Target-Trial Gate inside the Data Science Agent (`agents.py`). When the **Data Science Agent (DSA)** evaluates a behavioral or pharmacological contrast $\mathcal{T}$, it executes a three-condition identifiability check [10, 12, 13]: (1) propensity overlap $\text{Overlap}(\mathcal{T}) \geq 70.0\%$; (2) post-IPTW covariate balance $\max_m |\text{SMD}_m| \leq 0.12$; and (3) non-parametric Manski bounds $[\underline{\Delta}_{\text{Manski}}, \overline{\Delta}_{\text{Manski}}]$ alongside E-values. If positivity or balance fails, the gate returns `NOT_IDENTIFIABLE` (`risk_difference_point = None`).

Stage A4: *Drosophila* `240-KC` Value-of-Information (VOI) Router inside the Domain Expert Agent (`agents.py`). Inside the **Domain Expert Agent (DEA)**, every outpatient diagnostic channel j with epistemic status `MISSING_UNORDERED` or `PENDING_AT_LANDMARK` is scored for its Value of Information ($\text{VOI}_j \in [0, 1]$) and paired with an explicit counterfactual **`If-Wrong` risk interval** $[R_{\min}(t_k \mid j), R_{\max}(t_k \mid j)]$.

2.2 Direction B (`PHA` $\rightarrow$ `SALVAGE-AI`): Extending Anti-Reward-Hacking Triage into Home Coaching

Google's 3-agent PHA architecture extends `SALVAGE-AI` from a single acute hospital encounter into a continuous 90-day post-discharge recovery loop:

1. **Anti-Reward-Hacking Utility Alignment:** In `POST_DFI_SALVAGE`, planned minor ray resections that preserve the heel and plantar arch carry positive functional salvage utility (+3.00), directing coaching toward protective smart-boot offloading ($\geq$ 85% adherence, RD = -0.164). In `POST_SEPSIS_CARDIORENAL`, mild serum creatinine rise during oral loop diuresis with brisk natriuresis (UNa $>$ 70 mmol/L) and weight loss is recognized as **permissive hypercreatininemia** [14, 6] (+2.85 utility, RD = -0.131) rather than intrinsic tubular AKI.
2. **Fail-Closed `HealthCoachAgent` (`OARS` vs. Clinician `NowCard`):** On stable days (`ROUTINE_COACHING`), the HCA delivers personalized `OARS` Motivational Interviewing [2]. When escalation criteria trigger (`STAT_PROVIDER_CROSSROADS`), the HCA halts lifestyle coaching (`HALTED_FOR_CLINICIAN_ESCALATION`) and mints an HMAC-SHA256 signed **`sdr.decision.v1` Clinician NowCard**.

3 Experimental Design ($N = 160$)

We evaluated the bidirectional architecture on a deterministic benchmark of $N = 160$ **synthetic post-discharge patient trajectories** (`data/pha_salvage_bidirectional_benchmark.json`, generated by `src/pha_salvage_bridge/benchmark.py` with fixed seed `20261023`):

- **Cohort 1 (`POST_DFI_SALVAGE`, $n = 80$):** Post-debridement diabetic foot osteomyelitis/ulcer recovery monitored via smartwatch PPG, smart offloading boot adherence sensors, and a Bluetooth plantar thermometry mat ($\Delta T_{\text{plantar}}$), paired with outpatient wound cultures, serum CRP, HbA1c, and pedal Doppler hallux toe pressures.
- **Cohort 2 (`POST_SEPSIS_CARDIORENAL`, $n = 80$):** Post-sepsis cardiorenal recovery on oral loop diuretics monitored via smartwatch PPG and Bluetooth

Table 1: **Outpatient `charttime` vs. `storetime` Temporal Leakage Audit ($N = 160$ Trajectories at Day 7 Landmark $t_k = 168.0$h).**

Evaluation Protocol	Timestamp Filter Applied at $t_k = 168.0$h	30-Day Readmission AUROC	Leakage Inflation (ΔAUROC)	Delayed Outpatient Labs Masked (%)
Strict 4D State Lock (`PHA` × `SALVAGE-AI`)	$t_{\text{store}}(e_i) \leq 168.0$h	**0.8110**	Baseline (Zero Leakage)	**100.0%**
Naive Retrospective Join (`Leaked charttime`)	$t_{\text{chart}}(e_i) \leq 168.0$h	1.0000	+0.1890	0.0%

Table 2: **Three-Arm Architecture Comparison Across $N = 160$ Post-Discharge Inpatient-to-Home Trajectories ($n = 80$ `POST_DFI_SALVAGE` + $n = 80$ `POST_SEPSIS_CARDIORENAL`).**

Architectural Arm	Exposed to `charttime` Leakage?	Missing-Workup VOI Recall@3	Unidentifiable Causal Claims Blocked (%)	Unsafe Coaching During Acute Flare (%)
Arm 1: Monolithic Ungated LLM	Yes (Leaked)	0.4562 (45.6%)	0.0%	34.0%
Arm 2: Standard 3-Agent PHA (No `SALVAGE` Guards)	Yes (Leaked)	0.6875 (68.8%)	0.0%	12.5%
Arm 3: Bidirectional `PHA` × `SALVAGE-AI` Bridge	**No (`storetime` Locked)**	**1.0000 (100.0%)**	**100.0%**	**0.0%**

weight scales ($\Delta\text{Weight}_{72\text{h}}$), paired with outpatient serum creatinine (BMP), spot urine sodium (UNa), NT-proBNP, and point-of-care VExUS venous congestion grading.

4 Results

4.1 Outpatient `charttime` vs. `storetime` Leakage Audit

As shown in Table 1, across all $N = 160$ trajectories evaluated at Day 7 ($t_k = 168.0$h), 100.0% **of episodes** contained delayed outpatient lab or microbiology results ($t_{\text{chart}} \leq 168.0\text{h} < t_{\text{store}}$). Enforcing the strict 4D `storetime` lock prevented a +0.1890 **optimistic AUROC inflation** (0.8110 strict `storetime` vs. 1.0000 leaky `charttime`).

4.2 Three-Arm Architecture & Spatial Crossroads Audit

Table 2 compares the three architectural arms across all $N = 160$ episodes:

1. **Missing-Workup VOI Recall@3:** Arm 3 (`PHA` × `SALVAGE-AI`) achieved 1.0000 (100.0%) Top-3 missing-workup recall, outperforming both the monolithic LLM (0.4562) and the unguarded 3-agent PHA (0.6875).

2. **Target-Trial Positivity Gate:** Arm 3 blocked 100.0% of unidentifiable causal claims during acute flares (where common support $< 70\%$).

3. **Zero Unsafe Coaching During Acute Flares:** Arm 3 reduced unsafe lifestyle coaching during acute post-discharge flares from 34.0% (monolithic LLM) and 12.5% (unguarded PHA) to 0.0%.

4. **Apple Vision Pro (`visionOS 27.0`) Spatial Crossroads & `M0-M5` Symbiosis Gate:** Operating at 90.0 **Hz** with 0.68 **ms** gate latency and hard safety caps (Autonomy $\leq 40\%$, Deference $\geq 80\%$), the spatial gate suppressed 100.0% of routine benign wearable syncs (`PHA-02`, `PHA-04`, `PHA-06`) while achieving 100.0% wake sensitivity on clinical red flags (`PHA-01`, `PHA-05`; Table 3).

5 Discussion, Limitations, & Code Availability

Our results demonstrate that acute inpatient routers (`SALVAGE-AI`) [9] and multi-agent personal health systems [1] form complementary halves of a continuous hospital-to-home safety architecture. Prospective validation pairing continuous consumer wearables with real-world FHIR R4 `Observation.issued` (`storetime`) patient-portal feeds is the primary next clinical step.

Code & Data Availability. The zero-dependency Python 3.10+ reference implementation, $N = 160$ benchmark dataset, 6-test verification suite, and interactive Apple Vision Pro Spatial Crossroads console are openly available at https://github.com/deesatzed/personal-health-agent-bridge (companion to https://github.com/deesatzed/spatial-clinical-router).

Acknowledgments & Credits

This bidirectional architectural synthesis (`PHA` × `SALVAGE-AI`), $N = 160$ post-discharge inpatient-to-home benchmark, *Drosophila* `240-KC` sparse Value-of-Information router, and Apple Vision Pro (`visionOS 27.0`) Spatial Decision Crossroads & `M0-M5` Symbiosis Gate were developed with the support and collaborative engineering contributions of the **Google Science Program**.

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Table 3: **Six Flagship Post-Discharge Inpatient-to-Home Clinical Scenarios (`PHA-01`–`PHA-06`).**

ID	Cohort (t_k)	Clinical Scenario Summary	Vision Pro Spatial Gate Mode	DSA Target-Trial Gate (R_{30d})	DEA Top-1 VOI (`If-Wrong`) & HCA Mode
`PHA-01`	`POST_DFI` (168h)	+2.6°C plantar flare; CRP (172h) & wound swab (212h) pending (2 leaked rows blocked)	`HARD_ESCALATION` ($z = 3.06$)	`NOT_IDENTIFIABLE` (31.4% overlap; $R = 43.2\%$)	`pedal_doppler_toe_pressure` (VOI = 0.95, [0.238, 0.755]); `HALTED` (`NowCard`)
`PHA-02`	`POST_DFI` (336h)	Clean 2nd-ray resection remission; 94% boot adherence, $\Delta T = 0.5$°C, RHR 68 bpm	`SUPPRESSED_SYNC` ($z = 0.59$)	`EXPLORATORY_FEASIBLE` (RD = −0.164; $R = 11.9\%$)	`wound_probe_or_culture` (VOI = 0.92, [0.065, 0.432]); `ACTIVE_MI` (`OARS`)
`PHA-03`	`POST_DFI` (240h)	Calcified inpatient ABI = 1.42; unordered hallux toe pressure & subclinical $\Delta T = 1.8$°C drift	`STANDARD_REVIEW` ($z = 2.12$)	`EXPLORATORY_FEASIBLE` (RD = −0.164; $R = 23.6\%$)	`pedal_doppler_toe_pressure` (VOI = 0.95, [0.130, 0.559]); `ACTIVE_MI` (`OARS`)
`PHA-04`	`POST_SEPSIS` (120h)	−1.8 kg weight loss on torsemide; Cr 1.55 → 1.82 mg/dL with brisk UNa = 84 mmol/L	`SUPPRESSED_SYNC` ($z = 1.45$)	`EXPLORATORY_FEASIBLE` (RD = −0.131; $R = 5.7\%$)	`venous_pocus_vexus_grade` (VOI = 0.91, [0.031, 0.367]); `ACTIVE_MI` (`OARS`)
`PHA-05`	`POST_SEPSIS` (168h)	Acute +2.5 kg fluid surge in 72h, RHR 101 bpm, HRV 18 ms; BMP pending, UNa unordered	`HARD_ESCALATION` ($z = 3.85$)	`NOT_IDENTIFIABLE` (28.5% overlap; $R = 44.9\%$)	`serum_creatinine_mg_dl` (VOI = 0.97, [0.247, 0.779]); `HALTED` (`NowCard`)
`PHA-06`	`POST_SEPSIS` (240h)	Stable euvolemic recovery (ΔWeight = −0.1 kg, RHR 66 bpm, HRV 48 ms, Cr 0.95 mg/dL)	`SUPPRESSED_SYNC` ($z = 0.35$)	`EXPLORATORY_FEASIBLE` (RD = −0.131; $R = 10.4\%$)	`spot_urine_sodium_mmol_l` (VOI = 0.96, [0.057, 0.431]); `ACTIVE_MI` (`OARS`)

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